

MANET - I

Module 40

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about mobile ad hoc network first of all please see the contents which I shall cover after covering brief introduction about Manet I shall take up characteristics of Manet which will be followed by challenges of Manet networks in last I shall take up applications of mobile ad hoc network so now let us start like traditional wir networks wireless networks are formed by routers and hosts in a wireless network Network the routers are responsible for forwarding data packets in the network and hosts may be sources or syns of data flows the fundamental difference between

wired and wireless networks is the way that the network components communicate a wi Network relies on physical cables to transfer data in a wireless network the communication between different network components can be either wired or Wireless since wireless communication does not have the constraint of physical cables it allows a certain freedom for hosts and routers in a wireless network to move this is one of the advantages of a wireless network according to the mobility of hosts and routers there are three different types of wireless networks fixed wireless network fixed hosts and routers use wireless channels to communicate at with each other and hence form a fixed wireless network an example is a wireless network formed by fixed network devices using directed

antennas according to the figure two computers are communicating VI antenna signals through this they can send or receive the data wireless network with fixed exess points mobile hosts use wireless channels to communicate with fixed access points which may act as routers for those mobile hosts and to form a mobile network with fixed access points an example is the number of mobile laptop users in a building that access fixed access points according to the figure two or more laptops are sending or receiving the data via access points mobile ad hoc network a Manet is formed by mobile 3 minutes, 1 second hosts some of these mobile hosts are willing to forward packets for Neighbors examples include vehicle to vehicle and

ship to ship networks that communicate with each other by relying on peer-to-peer routings according to figure All Ships create a network without any centralized Administration mechanism they can easily establish the link with any node within the network it is an infrastructure less IP based network of mobile and wireless machine nodes connected with radio the nodes of a Manet do not have centralized Administration mechanism it is known for its route aware Network properties which each node act as a router to forward the traffic to other specific node in the network according to figure mobiles and laptops creates a network without any centralized Administration in the simplest scenarios nodes may be able to communicate directly with each other for example when they are within Wireless

transmission range of each other however ad hoc networks must also support communication between nodes that are only indirectly connected by a series of Wireless hops through other nodes according to figure to establish communication between nodes A and C the network must enlist the aid of node B to relay packets between them these circles indicate the nominal range of each nodes radio transceiver nodes A and C are not in direct transmission range of each other since a circle does not cover C characteristics of mobile ad hoc networks as compared to other wired or infrastructure based wireless networks manet has the following characteristics autonomous terminal distributed operation multi hop routing lightweight

terminals Dynamic topology self organization multi hopping resource conversation limited capacity and scalability autonomous terminal in manet each mobile terminal is an autonomous node which may function as a host or a router in other words beside the basic processing ability as a host the mobile nodes can also perform switching functions as a router so usually endpoints and switches are indistinguishable in Manet distributed operation since there is no background Network for the central control of the network operations the control and management of the network is distributed among the terminals the nodes involved in a Manet should collaborate Among Us themselves and each node acts as a relay

as needed to implement functions like security and routing multi hop routing in multihop routing ad hoc routing algorithms can be single hop or multi hop they are based on different link layer attributes and routing protocols single hop Manet is simpler than multi hop in terms of structure and implementation while transferring data packets from a source to its destination out of the direct Wireless transmission range the packets should be forwarded by intermediate nodes lightweight terminals in most of the cases the Manet nodes are mobile devices with less CPU processing capability small memory size and low power storage such devices need optimized algorithms and mechanisms that implement the Computing and communicating

functions Dynamic topology all nodes of Manet are free to move causing Network topology change rapidly at unpredictable times links between nodes are expected to break much more frequently than with wired and infrastructure based wireless networks self organization due to the lack of infrastructure or Central Administration nodes should be able to form themselves

into a network multi hopping in a Manet nodes use a wireless channel to transmit data and due to the limited number of nodes neighbors intermediate nodes are used to relay the packets resource conservation in manner the nodes are limited in both energy Supply and processing power power conservation becomes a very important factor to be considered when designing a network therefore optimizing

all operations May minimize the energy consumption limited security mobile ad hoc networks are more prone to security threats than wi networks or infrastructure based wireless networks because of their unique characteristics each mobile node in an ad hoc network and function as a router or packet forwarded for other nodes both legitimate users and malicious attackers can access the wireless Channel and there is no well place where excess control mechanisms can be deployed scalability in some applications mobile ad hoc networks May grow up to several thousand nodes mobile ad hoc networks suffer from scalability problems in Channel capacity because Channel capacities are very limited and maximum use of Channel capacity can be reached

9 minutes, 1 second faster due to the multi- hopping nature of mobile ad hoc networks their scalability is related to the routing protocols they employ low bandwidth these networks have lower capacity and shorter transmission range than fixed infrastructure networks the throughput of wireless communication is lesser than wir communication because of the effect of multiple excess fading noise and and interference conditions challenges of mobile ad hoc networks in last few years mobile ad hoc networking has been a popular field of communication almost every aspect of the network has been explored in one way or other at different level of problem the topics that need to be resolved are as follows scalability routing quality of service

client server model model shift security energy conservation node cooperation interoperation scalability most of The Visionaries depicting applications which are anticipated to benefit from the adhoc technology take scalability as granted imagine for example the vision of universal Computing where networks can be of any size however it is unclear how such large networks can actually grow adhoc networks suffer by Nature from the scalability problems in capacity routing routing in Wireless ad hoc networks is non-trivial due to the highly Dynamic environment an ad hoc network is a collection of Wireless Mobile nodes dynamically forming a temporary Network without the use of any pre-existing Network infrastructure or centralized Administration

in a typical ad hoc network mobile nodes come together for a period of time to exchange information while exchanging information the nodes may continue to move and so the network must be prepared to adapt continually to establish Roots among themselves without any outside support quality of service the heterogeneity of existing internet applications has challenged Network designers who have built the network to provide best effort service only Voice live video and file transfer are just a few applications having vast and diverse

requirements quality of service that is Q aware Solutions are being developed to meet the emerging requirements of these applications qos has to be guaranteed by the network to provide certain performance for a given flow or or a

collection of flows in terms of Qs parameters such as delay Jitter bandwidth packet loss probability and so on despite the current research efforts in Qs areas Qs in ad hoc networks is still an unexplored area client server model in the internet a network client is typically configur to use a server as its partner for Network transactions the these servers can be found automatically or by a static configuration in and hoc networks however the network structure cannot be defined by collecting IP addresses into subnets there may not be servers but the demand for basic Services still exists address allocation name resolution authentication and the service location itself are just examples of basic Services which are needed but their location in the network

is unknown and dynamically changes over time due to the infrastructure less nature of these networks and node Mobility a different addressing approach may be required in addition it is still not clear who will be responsible for managing various Network Services therefore while there have been vast research initiatives in this area the issue for shifting from the traditional client server model remains to be appropriately addressed security a vital issue that has to be addressed is the security in ad hoc networks applications like military and confidential meetings require High degree of security against enemies and active passive attackers ad hoc networks are particularly prone to malicious behavior lack of any centralized Network management or certification Authority

makes these dynamically changing Wireless structures security is often considered to be the major Rod Block in commercial application interoperation when two autonomous ad hoc networks move into same area the interference with each other becomes unavoidable and the networks would recognize the situation and be merged however the issue of joining two networks is not trivial the networks may be used using different synchronization or even different Mac or routing protocols security also becomes a major concern can the networks adapt to the situation for example a military unit moving into an area covered by a sensor Network could be such a situation moving unit would probably be using different routing protocol with location

information support while the sensor Network might be using a simple static router protocol so now we move on to the applications of ad hoc networks mobile ad hoc networks are best suitable for situations where an infrastructure is not available or to deploy one is not cost effective the following are some of the important applications military applications mobile ad hoc networks satisfies several military needs such as Battlefield survivability in such environment setting up of an infrastructure for communication between shoulders in Battlefield could be impossible the wireless devices carried by shoulders can form a mobile ad hoc network to support communication among them according to the figure shoulders tanks

helicopter and other vehicles carry the

mobile devices for message passing process home office and educational applications ad hoc networks also have applications in Home and Office environments the simplest and most direct application of ad hoc networks in both homes and offices is the networking of laptops pdas and other WLAN enable devices in the absence of Wireless based station another home application that falls within the personal area network that is pan class as wire replacement through Wireless links and as in Bluetooth all periphery devices can connect to computer through wireless Bluetooth links eliminating the need for wide connections ad hoc networks can also enable streaming of video and audio among Wireless nodes even in the absence of any base station for example students

in a classroom can use their laptops to obtain the latest class material from a professor's laptop as the class progresses universities and campus settings virtual classrooms ad hoc Communications during meetings or lectures are some of the educational applications of ad hoc networks a vehicular ad hoc network ret vehicular ad hoc networks are responsible for the communication between moving vehicles in a certain environment a vehicle can communicate with another vehicle directly which is called vehicle to vehicle that is v2v communication or a vehicle can communicate to an infrastructure such as roadside unit RSU known as vehicle to infrastructure v2i according to figure vanet scenario is creating VI v2v communication and v2i

communication between the vehicles on the road wireless sensor networks wireless sensor networks can be considered as special case of mobile ad hoc networks Manet with reduced or no Mobility initially wsn was mainly motivated by military applications on the civilian application domain of wireless sensor networks such as environmental and spaces monitoring disaster management smart home production and Healthcare Etc these wsn May consist of heterogeneous and mobile sensor nodes the Network topology may be as simple as start topology the scale and density of a network varies depending on the application according to the figure application of wireless sensor networks include disaster management Network

management home networks Healthcare informatics and Military applications sensor networks have emerged as a promising tool for monitoring the physical world utilizing self organization networks of battery powered Wireless sensors that can sense process and communicate a sensor network is a network of many tiny disposable lower part devices called nodes which are specially distributed in order to perform an application oriented Global task these nodes form a network by communicating with each other either directly or through other nodes one or more nodes among them will serve as sync that are capable of communicating with the user either directly or through the existing wild networks the primary component of the network is the sensor essential for monitoring real world physical

conditions such as sound temperature humidity vibration pressure motion pollutant Etc at different locations wireless mesh networks wireless mesh networks can easily effectively and wirelessly connect entire cities using inexpensive existing technology traditional networks rely on a small number of wired access points or wireless hotspots to connect users in a wireless mesh Network the network connection is spread out among dozens or even hundreds of wireless mesh nodes that talk to each other to share the network connection across a large area mesh nodes are small radio transmitters that function in the same way as wireless router nodes use the common Wi-Fi standards known as 802.11 a B and G to communicate

wirelessly with users and with each other it works on 2.4 GHz and 5 GHz frequency bands depending on the physical layer used for example if 802.11a is used the speed can be up to 54 Mbps nodes are programmed with software that tells them how to interact with a large Network information travels across the network from point A to point B by hopping wirelessly from one mesh node to the next the nodes automatically choose the quickest and safest path using a process known as Dynamic routing according to the figure there is an internet Cloud which is the backbone of mesh routers mesh routers are connected with access points base stations and other network devices all network devices provide the

internet access functionality to all connected users in the network disaster relief operations every year natural disasters like earthquake flood destroy people's lives around the world as the importance of the internet grows the loss of network connectivity during such disasters will be more noticeable effect of The Misfortune so it is important to find ways to enable the operations of networks even when infrastructure elements are disabled as a result of disasters personal area networks a personal area network that is pan is a computer network used for data transmission among devices such as computers telephones and personal digital assistants the idea of a personal area network is to create a network that

consists of nodes which are associated with a single person these nodes may be placed in a person's clothes belt or carried in handbags this type of network provides great flexibility for example it allows you to send a document to the printer in the office upstairs while you are sitting in the couch with your laptop using pan you can upload the photo from your cell phone to your desktop computer so friends this was all about Manet hope the concepts explained in this lecture were understandable and helpful in the next next lecture of Manet I will cover the routing protocols used in mobile ad hoc network hope to see you in the next lecture till then goodbye thank you